Notes: La Porta, Lopez-de-Silanes, Shleifer, & Vishny (JF, 1997)

Legal Determinants of External Finance

Contents

1 Big picture

- **Question.** Why are external capital markets much larger in some countries than in others?
- **Central hypothesis.** Better investor protection - legal rules plus enforcement - makes outside investors more willing to provide funds, so countries with stronger legal protection should have broader and more valuable equity and debt markets.
- **Empirical setting.** The paper studies a cross-section of 49 countries and classifies them by legal origin: English common law, French civil law, German civil law, and Scandinavian civil law.
- **Main institutional result.** Common law countries have stronger shareholder and creditor protections than civil law countries. French civil law countries have the weakest investor protections and the least developed capital markets.
- **Equity-market result.** Countries with stronger shareholder rights, better rule of law, and common-law legal origin have larger outsider-held stock market capitalization, more domestic listed firms per capita, and more IPOs per capita.
- **Debt-market result.** Debt/GNP is also larger when law enforcement is stronger. Creditor-rights measures are less powerful than shareholder-rights measures, but legal origin and enforcement still matter.
- **Firm-level result.** Large firms in weak legal systems still obtain external finance more easily than small firms. Aggregate market measures differ more sharply across legal origins than WorldScope large-firm medians.
- **Economic mechanism.** Legal protection reduces expropriation risk from insiders. Lower expropriation risk improves the terms on which entrepreneurs can sell equity or borrow, thereby expanding external finance.
- **Broad takeaway.** Finance is not determined only by technology, growth, or macroeconomic size. The legal environment is a central determinant of financial development.

2 Data and measurement

2.1 Country sample and legal origins

- The sample contains 49 countries.
- The paper assigns each country to the legal origin of its Company Law or Commercial Code: English, French, German, or Scandinavian.
- Legal origin is treated as a stable institutional source of variation because many countries inherited their systems through conquest, colonization, or legal transplantation.
- The main comparison is between common law and civil law families, especially English common law versus French civil law.

Interpretation. Legal origin is not a policy variable that can be changed overnight. It is used as a historical institutional classification that organizes differences in investor protections.

2.2 Aggregate equity finance measures

The paper uses three aggregate measures of equity-market development:

- **External cap/GNP:**

$$\frac{\text{stock market capitalization held by minority outside shareholders}}{\text{GNP}}.$$

- **Domestic firms/Pop:** number of domestic listed firms divided by population in millions.
- **IPOs/Pop:** number of initial public offerings between July 1995 and June 1996 divided by population in millions.

The external capitalization measure adjusts total market capitalization by removing the estimated fraction held by insiders:

$$\text{External cap} = extMarketcap \times (1 - \text{top-three shareholder ownership share}).$$

Interpretation. The adjustment is important because total market capitalization overstates true outside financing when insiders retain large controlling stakes.

2.3 Aggregate debt finance measure

The aggregate debt measure is

$$\frac{\text{private-sector bank debt} + \text{outstanding nonfinancial bonds}}{\text{GNP}}.$$

Interpretation. This measure captures broad access to debt finance, including bank lending and bond market borrowing. It is less focused on public firms than equity-market capitalization.

2.4 Legal protection and enforcement variables

- **Rule of law:** assessment of the country's law-and-order tradition, scaled from 0 to 10.
- **Antidirector rights:** shareholder-rights index from 0 to 5. It includes proxy voting by mail, no share-blocking before meetings, cumulative voting, oppressed-minority protection, and low thresholds for calling extraordinary meetings.
- **One-share = one-vote:** indicator equal to one if ordinary shares must carry one vote per share and firms cannot use multiple-vote or nonvoting ordinary shares.
- **Creditor rights:** index from 0 to 4 based on bankruptcy and reorganization protections for secured creditors.

Interpretation. The paper separates legal rules from law enforcement. Strong formal rights are less valuable if courts and enforcement institutions are weak.

2.5 Controls

- **GDP growth:** average annual per-capita GDP growth from 1970 to 1993.
- **Log GNP:** log of gross national product in 1994.

Interpretation. Growth controls for investment opportunities and valuation; log GNP controls for the scale of the economy. The paper generally avoids GDP per capita because it is highly correlated with rule of law.

2.6 Firm-level WorldScope measures

For the largest listed firms in 38 countries, the paper also studies:

- market capitalization held by minorities divided by sales;
- market capitalization held by minorities divided by cash flow;

- total debt divided by sales;
- total debt divided by cash flow;
- WorldScope coverage as a fraction of domestic listed firms.

Interpretation. These variables ask whether the aggregate legal-finance relation still appears when focusing on large firms that may have special access to international capital markets, banks, or governments.

3 Models and empirical specifications

3.1 The paper's core economic logic

The mechanism can be summarized as:

Better legal protection ⇒ lower expropriation risk ⇒ better terms for outside investors ⇒ more external finance.

Interpretation. Entrepreneurs can sell claims more easily when investors believe legal rules and courts protect them against diversion of cash flows by insiders.

3.2 Country-level external finance regressions

The main regression form is

$$Y_i = \alpha + \beta_1 GDPGrowth_i + \beta_2 \log(GNP_i) + \beta_3 RuleOfLaw_i + \delta' LegalOrigin_i + \gamma' Rights_i + \varepsilon_i, \quad (1)$$

where Y_i is one of the aggregate external-finance measures:

$$Y_i \in \{ExternalCap/GNP_i,\ DomesticFirms/Pop_i,\ IPOs/Pop_i,\ Debt/GNP_i\}.$$

Interpretation. The regression asks whether legal rules and enforcement predict financial development after controlling for growth and country size.

3.3 Legal-origin specifications

Legal-origin dummies are included as

$$LegalOrigin_i = (French_i, German_i, Scandinavian_i),$$

with English common law omitted.

Interpretation. Negative coefficients on civil-law dummies mean that civil-law countries have smaller capital markets than English-origin countries, conditional on controls.

3.4 Shareholder-rights specifications

For equity outcomes, the paper alternates between:

- the antidirector-rights index;
- the one-share-one-vote dummy;
- legal-origin dummies;
- combinations of legal origin and shareholder-rights measures.

Interpretation. If shareholder-rights coefficients remain positive, formal shareholder protections help explain equity-market development. If legal-origin coefficients remain negative, legal origin captures institutional features beyond the rights index.

3.5 Creditor-rights specification

For debt outcomes, the analogous specification uses creditor rights:

$$Debt/GNP_i = \alpha + \beta_1 GDPGrowth_i + \beta_2 \log(GNP_i) + \beta_3 RuleOfLaw_i + \delta' LegalOrigin_i + \gamma CreditorRights_i + \varepsilon_i. \quad (2)$$

Interpretation. Creditor-rights rules should matter for debt markets, but enforcement may be even more central because creditor claims depend heavily on courts and bankruptcy procedures.

3.6 Rights-sorted means

The paper also sorts countries into bottom, middle, and top groups by legal-rights variables and compares external-finance outcomes:

$$\bar{Y}_{Top25} - \bar{Y}_{Bottom25}.$$

Interpretation. These tests are less structural than regressions but give an intuitive picture: countries with stronger rights or enforcement tend to have broader markets.

3.7 Firm-level comparison

For the WorldScope firm-level sample, the paper compares country medians by legal origin and by aggregate market-development groups.

Interpretation. The firm-level exercise is a robustness and interpretation check. It shows whether the legal environment affects all firms or whether large firms can bypass weak domestic legal systems.

4 Identification and empirical design

4.1 What identifies the main association

- Identification comes from cross-country variation in legal origin, legal rules, law enforcement, and financial market development.
- The basic empirical claim is not a randomized causal design. It is a structured cross-country association motivated by legal history and investor-protection theory.
- Legal origin is useful because it predates the modern financial markets the paper studies and varies systematically across countries.

Interpretation. The paper's strongest evidence is that multiple financial-development measures line up with multiple legal-protection measures in a consistent way.

4.2 Why legal origin matters empirically

- English common law countries generally score higher on shareholder protection.
- French civil law countries score lower on shareholder and creditor protection.
- German and Scandinavian civil law countries are intermediate in many measures and often have stronger enforcement than French-origin countries.

Interpretation. Legal origin helps organize legal variation, but it may also proxy for deeper institutional, political, or cultural differences.

4.3 Why rule of law matters

- Rule of law is strongly associated with the breadth and valuation of both equity and debt markets.
- Moving from the sample-average rule-of-law score to a perfect 10 is economically large in the regressions.
- The paper interprets enforcement as a key complement to investor rights.

Interpretation. Investor rights on paper are less useful without reliable courts and enforcement.

4.4 Main credibility strengths

- The paper uses multiple external-finance measures rather than one market-size proxy.
- It separates equity and debt finance.
- It distinguishes formal legal rights from law enforcement.
- It uses legal origin to group countries by legal tradition rather than by current policy alone.
- It supplements aggregate evidence with WorldScope firm-level evidence.

4.5 What the paper cannot fully prove

- The evidence is cross-sectional and observational.
- Legal origin may be correlated with omitted institutions such as political history, trust, state capacity, colonial history, or culture.
- The sample is small, so estimates can be sensitive to influential countries.
- The main market variables are mostly from the mid-1990s.
- WorldScope covers mainly large listed firms and does not represent small private firms.
- The paper does not estimate the real effects of financial development on investment and growth.

Interpretation. The paper establishes a powerful legal-finance correlation and a persuasive mechanism, but it does not by itself settle all causal channels.

5 Tables and figures

The paper has no standalone figures. The compact notes therefore group the original tables and table panels directly cropped from the paper.

5.1 Table I: Variable definitions

Read:

- Table I defines the paper's central variables: legal origin, external cap/GNP, domestic firms/pop, IPOs/pop, debt/GNP, GDP growth, log GNP, rule of law, antidirector rights, one-share-one-vote, creditor rights, and WorldScope firm-level measures.
- The equity capitalization measure explicitly adjusts for insider ownership by using the fraction not owned by the top three shareholders.

- Antidirector rights ranges from 0 to 5; creditor rights ranges from 0 to 4; rule of law ranges from 0 to 10.

Economic message. Measurement is central. The paper is not merely comparing total stock market values; it tries to measure external finance available to outside investors.

Table I

Description of the Variables

Origin	Identifies the legal origin of the Company Law or Commercial Code of each country. Source: Reynolds and Flores (1989) and La Porta *et al*. (1996).
External cap/ GNP	The ratio of the stock market capitalization held by minorities to gross national product for 1994. The stock market capitalization held by minorities is computed as the product of the aggregate stock market capitalization and the average percentage of common shares not owned by the top three shareholders in the ten largest non-financial, privately-owned domestic firms in a given country. A firm is considered privately owned if the State is not a known shareholder in it. Source: *Moodys International*, *CIFAR*, *EXTEL*, *WorldScope*, *20-Fs*, *Price-Waterhouse*, and various country sources.
Domestic firms/ Pop	Ratio of the number of domestic firms listed in a given country to its population (in millions) in 1994. Source: *Emerging Market Factbook* and *World Development Report 1996*.
IPOs/Pop	Ratio of the number of initial public offerings of equity in a given country to its population (in millions) for the period 1995:7–1996:6. Source: *Securities Data Corporation*, *AsiaMoney*, *LatinFinance*, *GT Guide to World Equity Markets*, and *World Development Report 1996*.
Debt/GNP	Ratio of the sum of bank debt of the private sector and outstanding non-financial bonds to GNP in 1994, or last available. Source: *International Financial Statistics*, *World Bondmarket Factbook*.
GDP growth	Average annual percent growth of per capita gross domestic product for the period 1970–1993. Source: *World Development Report 1995*.
Log GNP	Logarithm of the Gross National Product in 1994. Source: *World Development Report 1996*.
Rule of law	Assessment of the law and order tradition in the country. Average of the months of April and October of the monthly index between 1982 and 1995. Scale from 0 to 10, with lower scores for less tradition for law and order. Source: *International Country Risk Guide*.
Antidirector rights	An index aggregating shareholder rights. The index is formed by adding 1 when: (1) the country allows shareholders to mail their proxy vote; (2) shareholders are not required to deposit their shares prior to the General Shareholders' Meeting; (3) cumulative voting is allowed; (4) an oppressed minorities mechanism is in place; or (5) when the minimum percentage of share capital that entitles a shareholder to call for an Extraordinary Shareholders' Meeting is less than or equal to 10% (the sample median). The index ranges from 0 to 5. Source: Company Law or Commercial Code and La Porta *et al*. (1996).
One-share = one-vote	Equals one if the Company Law or Commercial Code of the country requires that ordinary shares carry one vote per share, and 0 otherwise. Equivalently, this variable equals one when the law prohibits the existence of both multiple-voting and non-voting ordinary shares and does not allow firms to set a maximum number of votes per shareholder irrespective of the number of shares she owns, and 0 otherwise. Source: Company Law or Commercial Code and La Porta *et al*. (1996).

(a) Table I, part 1

Table I—Continued

Creditor rights	An index aggregating creditor rights. The index is formed by adding 1 when: (1) the country imposes restrictions, such as creditors' consent or minimum dividends, to file for reorganization; (2) secured creditors are able to gain possession of their security once the reorganization petition has been approved (no automatic stay); (3) the debtor does not retain the administration of its property pending the resolution of the reorganization; (4) secured creditors are ranked first in the distribution of the proceeds that result from the disposition of the assets of a bankrupt firm. The index ranges from 0 to 4. Source: Company Law or Bankruptcy Laws and La Porta *et al*. (1996).
Market cap/ sales	The median ratio of the stock market capitalization held by minorities to sales in 1994 for all nonfinancial firms in a given country on the *WorldScope* database. Firm's *j* stock market capitalization held by minorities is computed as the product of the stock market capitalization of firm *j* and the average percentage of common shares not owned by the top three shareholders in the ten largest nonfinancial, privately-owned domestic firms in a given country. A firm is considered privately owned if the State is not a known shareholder in it. Source: *WorldScope*.
Market cap/ cash-flow	The median ratio of the stock market capitalization held by minorities to cash flow in 1994 for all nonfinancial firms in a given country on the *WorldScope* database. Firm's *j* stock market capitalization held by minorities is computed as the product of the stock market capitalization of firm *j* and the average percentage of common shares not owned by the top three shareholders in the ten largest nonfinancial, privately-owned domestic firms in a given country. A firm is considered privately owned if the State is not a known shareholder in it. Source: *WorldScope*.
Debt/sales	Median of the total-debt-to-sales ratio in 1994 for all firms in a given country on the *WorldScope* database. Source: *WorldScope*.
Debt/cash flow	Median of the total-debt-to-cash-flow ratio for all firms in a given country on the *WorldScope* database. Source: *WorldScope*.

(b) Table I, continued

5.2 Table II: External capital markets by legal origin

Read:

- English-origin countries have the largest average external cap/GNP: 0.60 versus 0.21 for French-origin countries.
- English-origin countries have more domestic listed firms per million people: 35.45 versus 10.00 for French-origin countries.
- English-origin countries have more IPOs per million people: 2.23 versus 0.19 for French-origin countries.
- English-origin countries score higher on antidirector rights: 3.39 versus 1.76 for French-origin countries.
- German-origin countries have high debt/GNP and high rule-of-law scores, complicating a simple common-law-versus-civil-law interpretation for debt.

- Panel B shows statistically strong differences for several comparisons, especially English versus French origin in equity-market breadth and shareholder rights.

Economic message. The raw cross-country means show the paper's core pattern: legal families with stronger investor protection tend to have broader and more valuable external capital markets.

Table II

External Capital Markets

This table classifies countries by legal origin. Definitions for each of the variables can be found in Table I. Panel B reports tests of means for the different legal origins.

Country	External Cap/GNP	Domestic Firms/Pop	IPOs/Pop	Debt/GNP	GDP growth	Log GNP	Rule of Law	Antidirector Rights	One-Share = One-Vote	Creditor Rights
				Panel A: Means						
Australia	0.49	63.55	–	0.76	3.06	12.64	10.00	4	0	1
Canada	0.39	40.86	4.93	0.72	3.36	13.26	10.00	4	0	1
Hong Kong	1.18	88.16	5.16	–	7.57	11.56	8.22	4	1	4
India	0.31	7.79	1.24	0.29	4.34	12.50	4.17	2	0	4
Ireland	0.27	20.00	0.75	0.38	4.25	10.73	7.80	3	0	1
Israel	0.25	127.60	1.80	0.66	4.39	11.19	4.82	3	0	4
Kenya	–	2.24	–	–	4.79	8.83	5.42	3	0	4
Malaysia	1.48	25.15	2.89	0.84	6.90	11.00	6.78	3	1	4
New Zealand	0.28	69.00	0.66	0.90	1.67	10.69	10.00	4	0	3
Nigeria	0.27	1.68	–	–	3.43	10.36	2.73	3	0	4
Pakistan	0.18	5.88	–	0.27	5.50	10.88	3.03	4	1	4
Singapore	1.18	80.00	5.67	0.60	1.68	11.68	8.57	3	1	3
South Africa	1.45	16.00	0.05	0.93	7.48	10.92	4.42	4	0	4
Sri Lanka	0.11	11.94	0.11	0.25	4.04	9.28	1.90	2	0	3
Thailand	0.56	6.70	0.56	0.93	7.70	11.72	6.25	3	0	3
UK	1.00	35.68	2.01	1.13	2.27	13.86	8.57	4	0	4
US	0.58	30.11	3.11	0.81	2.74	15.67	10.00	5	0	1
Zimbabwe	0.18	5.81	–	–	2.17	8.63	3.68	3	0	4
English origin avg	**0.60**	**35.45**	**2.23**	**0.68**	**4.30**	**11.41**	**6.46**	**3.39**	**0.22**	**3.11**
Argentina	0.07	4.58	0.20	0.19	1.40	12.40	5.35	4	0	1
Belgium	0.17	15.50	0.30	0.38	2.46	12.29	10.00	0	0	2
Brazil	0.18	3.48	0.00	0.39	3.95	13.03	6.32	3	1	1
Chile	0.80	19.92	0.35	0.63	3.35	10.69	7.02	3	1	2
Colombia	0.14	3.13	0.05	0.19	4.38	10.82	2.08	1	0	0
Ecuador	–	13.18	0.09	–	4.55	9.49	6.67	2	0	4
Egypt	0.08	3.48	–	–	6.13	10.53	4.17	2	0	4
France	0.23	8.05	0.17	0.96	2.54	14.07	8.98	2	0	0
Greece	0.07	21.60	0.30	0.23	2.46	11.25	6.18	1	1	1
Indonesia	0.15	1.15	0.10	0.42	6.38	11.84	3.98	2	0	4
Italy	0.08	3.91	0.31	0.55	2.82	13.94	8.33	0	0	2
Jordan	–	23.75	–	0.70	1.20	8.49	4.35	1	0	–
Mexico	0.22	2.28	0.03	0.47	3.07	12.69	5.35	0	0	0
Netherlands	0.52	21.13	0.66	1.08	2.55	12.68	10.00	2	0	2
Peru	0.40	9.47	0.13	0.27	2.82	10.92	2.50	2	1	0
Philippines	0.10	2.90	0.27	0.10	0.30	10.44	2.73	4	0	0
Portugal	0.08	19.50	0.50	0.64	3.52	11.41	8.68	2	0	1
Spain	0.17	9.71	0.07	0.75	3.27	13.19	7.80	2	0	2
Turkey	0.18	2.93	0.05	0.15	5.05	12.08	5.18	2	0	2
Uruguay	–	7.00	0.00	0.26	1.96	9.40	5.00	1	1	2
Venezuela	0.08	4.28	0.00	0.10	2.65	10.99	6.37	1	0	–
French origin avg	**0.21**	**10.00**	**0.19**	**0.45**	**3.18**	**11.55**	**6.05**	**1.76**	**0.24**	**1.58**
Austria	0.06	13.87	0.25	0.79	2.74	12.13	10.00	2	0	3
Germany	0.13	5.14	0.08	1.12	2.60	14.46	9.23	1	0	3
Japan	0.62	17.78	0.26	1.22	4.13	15.18	8.98	3	1	2
South Korea	0.44	15.88	0.02	0.74	9.52	12.73	5.35	2	1	3
Switzerland	0.62	33.85	–	–	1.18	12.44	10.00	1	0	1
Taiwan	0.88	14.22	0.00	–	11.56	12.34	8.52	3	0	2
German origin avg	**0.46**	**16.79**	**0.12**	**0.97**	**5.29**	**13.21**	**8.68**	**2.00**	**0.33**	**2.33**
Denmark	0.21	50.40	1.80	0.34	2.09	11.84	10.00	3	0	3
Finland	0.25	13.00	0.60	0.75	2.40	11.49	10.00	2	0	1
Norway	0.22	33.00	4.50	0.64	3.43	11.62	10.00	3	0	2
Sweden	0.51	12.66	1.66	0.55	1.79	12.28	10.00	2	0	2
Scandinavian origin avg	**0.30**	**27.26**	**2.14**	**0.57**	**2.42**	**11.80**	**10.00**	**2.50**	**0.00**	**2.00**
Sample average	**0.40**	**21.59**	**1.02**	**0.59**	**3.79**	**11.72**	**6.85**	**2.44**	**0.22**	**2.30**

Table 1: Table II, Panel A: external capital markets by legal origin

Table II—Continued

Country	External Cap/GNP	Domestic Firms/Pop	IPOs/Pop	Debt/GNP	GDP growth	Log GNP	Rule of Law	Antidirector Rights	One-Share = One-Vote	Creditor Rights
				Panel B: Tests of Means (t-statistics)						
Common vs civil law	3.12	3.16	3.97	1.33	1.23	−1.06	−0.77	5.24	−0.03	3.61
English vs French origin	3.29	3.16	4.50	2.29	1.97	−0.28	0.51	5.13	−0.11	3.61
English vs German origin	0.68	1.24	2.34	−1.88	−0.78	−2.31	−1.82	3.66	−0.52	1.43
English vs Scand. origin	1.25	0.44	0.08	0.71	1.81	−0.44	−15.57	2.14	2.20	1.71
French vs German origin	−2.38	−1.85	0.78	−3.39	−1.96	−2.48	−2.55	−0.47	−0.45	−1.29
French vs Scand. origin	−0.91	−3.31	−5.45	0.82	0.97	−0.33	−20.80	−1.25	2.50	−0.60
German vs Scand. origin	0.94	−1.21	−2.76	2.71	1.32	2.11	−11.29	−0.98	1.58	0.63

Table 2: Table II, Panel B: tests of means

5.3 Table III: Investor rights and external finance

Read:

- Countries in the top 25% of antidirector rights have external cap/GNP of 0.58, versus 0.19 for the bottom 25%.
- Countries in the top 25% of antidirector rights have 35.68 domestic firms per million people, versus 12.05 for the bottom 25%.
- One-share-one-vote countries have higher external cap/GNP, 0.65 versus 0.32.
- Countries in the top rule-of-law quartile have more listed firms, more IPOs, and higher debt/GNP than countries in the bottom quartile.
- GDP growth and log GNP matter, but they do not organize all outcomes as cleanly as investor protection and enforcement.

Economic message. The sorting results show that both formal rights and enforcement are linked to market breadth, not only to market valuation.

Table III

Investor Rights and External Finance

This table classifies countries according to their ranking in: (a) Antidirector Rights; (b) One-Share = One-Vote; (c) Creditor Rights; (4) Rule of Law; (5) GDP Growth; and (5) Log GNP. For each panel, the table shows the average value of different external finance measures for the bottom quartile, the middle two quartiles, and the top quartile. The last row of each panel shows the *t*-statistic for a test of means between the bottom and the top quartiles.

	External Cap/ GNP	Domestic Firms/Pop	IPOs/Pop	Debt/GNP
Means by antidirector rights				
Bottom 25%	0.19	12.05	0.14	0.44
Mid 50%	0.39	20.03	0.97	0.63
Top 25%	0.58	35.68	2.05	0.63
Test of means (*t*-statistic)				
Bottom 25% vs. Top 25%	−2.50	−2.35	−2.55	−1.22
Means by one-share = one-vote				
Not One Vote	0.32	20.10	0.87	0.59
One Vote	0.65	26.76	1.48	0.56
Test of means (*t*-statistic)				
One Vote vs Not One Vote	−2.61	−0.76	−1.08	0.29
Means by creditor rights				
Bottom 25%	0.27	18.43	0.85	0.49
Mid 50%	0.40	18.25	0.62	0.66
Top 25%	0.59	31.30	2.37	0.65
Test of means (*t*-statistic)				
Bottom 25% vs. Top 25%	−2.09	−1.11	−1.95	−1.15
Means by rule of law				
Bottom 25%	0.28	8.51	0.28	0.34
Mid 50%	0.47	22.36	0.89	0.63
Top 25%	0.36	33.08	1.85	0.70
Test of means (*t*-statistic)				
Bottom 25% vs. Top 25%	−0.73	−4.11	−2.30	−3.84
Means by GDP growth				
Bottom 25%	0.42	22.83	0.74	0.54
Mid 50%	0.28	15.90	0.86	0.60
Top 25%	0.62	30.43	1.64	0.62
Test of means (*t*-statistic)				
Bottom 25% vs. Top 25%	−1.05	−0.61	−1.20	−0.56
Means by log GNP				
Bottom 25%	0.25	15.36	0.27	0.43
Mid 50%	0.46	26.12	1.33	0.50
Top 25%	0.39	19.82	0.98	0.82
Test of means (*t*-statistic)				
Bottom 25% vs. Top 25%	−1.31	−0.63	−1.24	−3.26

Table 3: Table III: investor rights and external finance

5.4 Tables IV and V: Equity market capitalization and listed firms

Read: Table IV:

- Rule of law is positive and significant in most specifications for external cap/GNP.
- Antidirector rights is strongly positive when included alone: 0.1171.
- One-share-one-vote is positive when included alone: 0.2745.
- French, German, and Scandinavian origin dummies are negative relative to English origin.
- The adjusted R^2 rises to 0.3801 in the specification with one-share-one-vote and legal-origin controls.

Read: Table V:

- Rule of law is strongly positive for domestic firms per population in every specification.
- Antidirector rights is positive when included alone.
- Civil-law origin dummies are negative, often implying about 20 fewer listed firms per million people relative to English-origin countries.
- Log GNP is negative in several specifications, suggesting larger economies need not have more listed firms per capita.

Economic message. Shareholder protection and enforcement both matter for equity-market development. Legal origin still captures institutional differences beyond the specific rights indices.

Table IV

External Market Capitalization of Equity/GNP Regressions

Ordinary least squares regressions of the cross-section of 49 countries around the world. The dependent variable is "External Cap." The independent variables are (1) GDP Growth; (2) Log GNP; (3) Rule of law; (4) French origin; (5) German origin; (6) Scandinavian origin; (7) Antidirector Rights; (8) One-share = One-Vote. Standard errors are shown in parentheses.

Independent Variables	Dependent Variable: External Cap/GNP				
GDP growth	0.0617[b]	0.0544[b]	0.0584[b]	0.0562[b]	0.0441[b]
	(0.0232)	(0.0201)	(0.0238)	(0.0242)	(0.0209)
Log GNP	−0.0129	−0.0168	0.0038	−0.0053	0.0091
	(0.0333)	(0.0334)	(0.0386)	(0.0382)	(0.0324)
Rule of law	0.0378[c]	0.0455[b]	0.0417	0.0424[b]	0.0437[c]
	(0.0206)	(0.0203)	(0.0250)	(0.0243)	(0.0231)
French origin			−0.3225[a]	−0.2142[c]	−0.3341[a]
			(0.1131)	(0.1194)	(0.1084)
German origin			−0.2962[c]	−0.1849	−0.3230[b]
			(0.1497)	(0.1599)	(0.1438)
Scandinavian origin			−0.3391[b]	−0.2816[c]	−0.3056[b]
			(0.1373)	(0.1479)	(0.1218)
Antidirector rights	0.1171[a]			0.0675[c]	
	(0.0353)			(0.0354)	
One-share = one-vote		0.2745[b]			0.2890[b]
		(0.1235)			(0.1111)
Intercept	−0.2437	0.0100[b]	0.0336	−0.0860	−0.0475
	(0.2880)	(0.3063)	(0.3677)	(0.3629)	(0.3066)
Observations	45	45	45	45	45
Adjusted R^2	0.2936	0.2347	0.2867	0.3016	0.3801

(a) Table IV: External Cap/GNP regressions

Table V

Domestic Firms/Population Regressions

Ordinary least squares regressions of the cross-section of 49 countries around the world. The dependent variable is "Domestic Firms/Pop." The independent variables are (1) GDP growth; (2) Log GNP; (3) Rule of law; (4) French origin; (5) German origin; (6) Scandinavian origin; (7) Antidirector rights; (8) One-share = one-vote. Standard errors are shown in parentheses.

Independent Variables	Dependent Variable: Domestic Firms/Pop				
GDP growth	1.0767	1.3461	1.0111	0.8950	0.5763
	(1.4000)	(1.3318)	(1.2661)	(1.2733)	(0.9884)
Log GNP	−4.3181[b]	−4.0659[b]	−2.9126	−3.3073[b]	−2.7979
	(1.6588)	(1.7697)	(1.7698)	(1.8165)	(1.6816)
Rule of law	4.5093[a]	4.8584[a]	4.8422[a]	4.8577[a]	4.9582[a]
	(1.2579)	(1.4023)	(1.3616)	(1.3377)	(1.3356)
French origin			−21.9069[a]	−17.5313[b]	−22.5204[a]
			(7.4014)	(8.9183)	(7.2884)
German origin			−25.1485[a]	−20.5611[b]	−26.3007[a]
			(8.4882)	(9.7216)	(7.8639)
Scandinavian origin			−22.2680[b]	−19.9575[c]	−21.3009[b]
			(10.1744)	(10.0144)	(10.0541)
Antidirector rights	7.3034[a]			2.7304	
	(1.8052)			(1.6591)	
One-share = one-vote		8.1382			10.0675
		(7.5228)			(6.3165)
Intercept	19.3863	29.0780[c]	33.0485	28.6987	30.6212
	(15.4445)	(17.0108)	(20.6317)	(21.4015)	(20.2510)
Number of observations	49	49	49	49	49
Adjusted R^2	0.2198	0.1153	0.2197	0.2495	0.2681

[a] Significant at 1%; [b] Significant at 5%; [c] Significant at 10%.

(b) Table V: Domestic Firms/Pop regressions

5.5 Tables VI and VII: IPOs and debt markets

Read: Table VI:

- Rule of law is positive and significant for IPOs/pop across specifications.
- Antidirector rights is positive and significant.
- One-share-one-vote is positive and significant in the final specification.
- French and German origin countries have significantly fewer IPOs per capita than English-origin countries.
- Scandinavian origin is not significantly negative for IPOs.

Read: Table VII:

- Rule of law is positive and significant for debt/GNP in all specifications.

- Creditor rights is positive when included without legal-origin dummies, but weaker once origins are controlled.
- French-origin countries have lower debt/GNP in some specifications.
- Scandinavian-origin countries show lower debt/GNP relative to English-origin countries.

Economic message. Equity-market breadth is strongly related to shareholder protection. Debt-market depth appears especially tied to enforcement quality, while the creditor-rights index explains less of the origin differences.

Table VI

Initial Public Offerings/Population Regressions

Ordinary least squares regressions of the cross-section of 49 countries around the world. The dependent variable is "IPOs/Pop." The independent variables are (1) GDP growth; (2) Log GNP; (3) Rule of law; (4) French origin; (5) German origin; (6) Scandinavian origin; (7) Antidirector rights; (8) One-share = one-vote. Standard errors are shown in parentheses.

Independent Variables	Dependent Variable: IPOs/Pop				
GDP growth	0.1222	0.1320	0.1937^{b}	0.1916^{c}	0.1633^{b}
	(0.1281)	(0.1193)	(0.1012)	(0.1037)	(0.0744)
Log GNP	−0.1672	−0.1225	0.0662	0.0452	0.1255
	(0.1453)	(0.1692)	(0.1086)	(0.1129)	(0.1002)
Rule of law	0.2549^{a}	0.2943^{a}	0.2122^{b}	0.2108^{b}	0.2127^{a}
	(0.0889)	(0.0926)	(0.0842)	(0.0830)	(0.0731)
French origin			−1.5982^{a}	−1.2949^{a}	−1.6677^{a}
			(0.3552)	(0.3696)	(0.3132)
German origin			−2.8118^{a}	−2.5450^{a}	−3.027^{a}
			(0.5698)	(0.5909)	(0.5543)
Scandinavian origin			−0.3123	−0.1421	−0.1367
			(0.8666)	(0.8486)	(0.8414)
Antidirector rights	0.5352^{a}			0.1937^{c}	
	(0.1364)			(0.0989)	
One-share = one-vote		0.6359			1.0287^{a}
		(0.5422)			(0.3450)
Intercept	−0.5546	−0.2720	−0.9201	−1.3071	−1.7268
	(1.3472)	(1.7534)	(1.3233)	(1.3204)	(1.2088)
Number of observations	41	41	41	41	41
Adjusted R^2	0.3082	0.1571	0.4907	0.4927	0.5643

^{a} Significant at 1%; ^{b} Significant at 5%; ^{c} Significant at 10%.

(a) Table VI: IPOs/Pop regressions

Table VII

Debt/GNP Regressions

Ordinary least squares regressions of the cross-section of 49 countries around the world. The dependent variable is "Debt/GNP." The independent variables are (1) GDP growth; (2) Log GNP; (3) Rule of law; (4) French origin; (5) German origin; (6) Scandinavian origin; (7) Creditor rights. Standard errors are shown in parentheses.

Independent Variables	Dependent Variable: Debt/GNP		
GDP growth	0.0310^{c}	0.0251^{c}	0.0197
	(0.0171)	(0.0134)	(0.0152)
Log GNP	0.0667^{b}	0.0370	0.0404
	(0.0252)	(0.0255)	(0.0250)
Rule of law	0.0615^{a}	0.0698^{a}	0.0694^{a}
	(0.0132)	(0.0147)	(0.0148)
French origin		−0.1516^{b}	−0.1163
		(0.0740)	(0.0825)
German origin		0.1080	0.1082
		(0.1010)	(0.0982)
Scandinavian origin		−0.2764^{b}	−0.2618^{b}
		(0.1037)	(0.1075)
Creditor rights	0.0518^{c}		0.0270
	(0.0267)		(0.0298)
Intercept	−0.8621^{a}	−0.3496	−0.4414
	(0.2579)	(0.2524)	(1.341)
Number of observations	39	39	39
Adjusted R^2	0.5522	0.5191	0.5984

^{a} Significant at 1%; ^{b} Significant at 5%; ^{c} Significant at 10%.

(b) Table VII: Debt/GNP regressions

5.6 Table VIII: External funding at the firm level

Read:

- Table VIII studies 38 countries in WorldScope, focusing on large listed firms.
- English-origin countries have higher median market cap/sales than French and Scandinavian-origin countries, but the firm-level differences are weaker than aggregate differences.
- German-origin countries have high market cap/cash-flow values, partly because of high valuations in Japan and Taiwan.
- Debt ratios for large firms are much more similar across legal origins than aggregate debt/GNP measures.
- Countries with high aggregate external cap/GNP also have higher firm-level valuation ratios.
- Countries with low aggregate debt/GNP do not necessarily have low debt ratios for their largest public firms.

Economic message. The legal environment appears to matter most for broad access to finance. Large public firms can often obtain external debt even in weaker legal systems, possibly through banks, governments, or international capital markets.

Table VIII

External Funding at the Firm Level

The sample of thirty-eight countries includes all the firms on the Worldscope database for 1996. The table shows median values for all the firms in each country. Panel A show the medians based on a classification by legal origin. The definition for each of the variables can be found in Table I. Panel B gives the tests of means for the different legal origins. Panel C shows mean of medians and *t*-tests for countries sorted by levels of "External Cap/GNP." Panel D shows mean of medians and *t*-tests for countries sorted by "Debt/GNP."

Country	Market Cap/Sales	Market Cap/ Cash-Flow	Debt/ Sales	Debt/Cash-Flow	WorldScope Firms/ Domestic Firms
Panel A: Median Values by Legal Origin					
Australia	0.75	6.15	0.19	1.42	0.12
Canada	0.76	4.66	0.30	2.07	0.26
Hong Kong	0.66	4.01	0.31	2.50	0.12
India	0.73	8.75	0.47	4.26	0.01
Ireland	0.75	3.51	0.16	0.74	0.29
Israel	0.34	3.79	0.17	1.41	0.03
Malaysia	1.46	6.82	0.24	1.45	0.23
New Zealand	0.38	4.26	0.23	2.74	0.11
Pakistan	0.50	4.18	0.33	2.34	0.05
Singapore	0.83	5.68	0.07	0.83	0.19
South Africa	0.40	3.23	0.29	2.06	0.22
Thailand	0.71	4.65	0.54	3.45	0.32
UK	0.64	5.77	0.11	1.06	0.51
US	0.67	6.70	0.18	1.86	0.28
Average English origin	**0.69**	**5.16**	**0.26**	**2.01**	**0.20**
Argentina	0.63	4.18	0.28	1.78	0.10
Belgium	0.16	2.28	0.25	2.52	0.39
Brazil	0.24	1.97	0.18	1.52	0.11
Chile	1.68	8.15	0.29	1.59	0.13
France	0.29	4.28	0.19	2.36	0.67
Greece	0.25	5.99	0.21	2.55	0.04
Indonesia	0.48	3.03	0.37	3.25	0.23
Italy	0.17	2.21	0.32	3.04	0.44
Mexico	0.47	4.06	0.66	1.54	0.29
Netherlands	0.27	3.93	0.11	1.33	0.42
Philippines	1.61	5.17	0.29	0.86	0.14
Portugal	0.19	2.48	0.33	3.73	0.17
Spain	0.27	3.28	0.25	2.33	0.15
Turkey	0.46	2.87	0.11	0.50	0.12
Average French origin	**0.51**	**3.85**	**0.27**	**2.06**	**0.24**
Austria	0.21	2.29	0.24	2.38	0.17
Germany	0.21	3.29	0.10	1.24	0.55
Japan	0.63	13.80	0.34	6.99	0.50
South Korea	0.29	—	0.58	—	0.09
Switzerland	0.26	3.06	0.30	3.14	0.36
Taiwan	2.21	14.94	0.26	2.16	0.20
Average German origin	**0.63**	**7.48**	**0.30**	**3.18**	**0.31**
Denmark	0.30	3.30	0.22	1.88	0.38
Finland	0.30	2.90	0.31	2.58	0.80
Norway	0.49	3.70	0.36	3.62	0.46
Sweden	0.40	3.10	0.21	1.59	0.82
Average Scandinavian origin	**0.37**	**3.25**	**0.28**	**2.42**	**0.61**
Sample average	**0.58**	**4.77**	**0.27**	**2.24**	**0.28**

Table VIII—Continued

Country	Market Cap/Sales	Market Cap/ Cash-Flow	Debt/ Sales	Debt/Cash-Flow	WorldScope Firms/ Domestic Firms
Panel B: Tests of Means (*t*-statistics)					
Common vs. civil law	1.04	0.64	−0.60	−0.87	
England vs. France	1.10	2.11	−0.36	−0.14	
England vs. Germany	0.20	−1.33	−0.71	−1.61	
England vs. Scandinavia	2.17	2.38	−0.32	−0.73	
France vs. Germany	−0.42	−2.04	−0.43	−1.59	
France vs. Scandinavia	0.54	0.69	−0.06	−0.68	
Germany vs. Scandinavia	0.64	1.32	0.30	0.64	
Panel C: Sorted by External Cap/GNP			Panel D: Sorted by Debt/GNP		
Means					
Bottom 25%	0.29	3.23	0.26	1.94	
Mid 50%	0.53	4.31	0.29	2.17	
Top 25%	0.97	7.28	0.24	2.52	
Test of means					
Bottom 25% vs. Top 25%	−3.03	−2.68	0.33	−0.80	

6 Short summary

- The paper studies whether legal rules and law enforcement determine access to external finance across 49 countries.
- It builds on the idea that investors provide external capital only if they are protected from expropriation by insiders.
- Countries are classified by legal origin: English, French, German, and Scandinavian.
- The main equity outcomes are outsider-held stock market capitalization/GNP, listed domestic firms per population, and IPOs per population.
- The main debt outcome is private-sector bank debt plus corporate bonds divided by GNP.
- The main legal variables are rule of law, antidirector rights, one-share-one-vote, and creditor rights.
- English common law countries have the strongest investor protections and the broadest equity markets.
- French civil law countries have the weakest investor protections and the least developed capital markets.
- Rule of law is strongly associated with both equity and debt market development.
- Antidirector rights and one-share-one-vote rules help explain equity-market development.
- Creditor rights matter less robustly than shareholder rights, especially once legal origin is controlled.
- Legal-origin dummies remain important even after including specific investor-protection measures.

- Firm-level WorldScope evidence shows that large firms can obtain external finance more easily than the aggregate evidence suggests.
- The paper does not claim a randomized causal design; it documents strong and systematic legal-finance correlations.
- The main contribution is to make law and investor protection central to the study of financial development.

7 Conclusion

7.1 Core conclusion

La Porta, Lopez-de-Silanes, Shleifer, and Vishny show that the legal environment is a major determinant of external finance. Countries with stronger investor protections and better enforcement have larger and broader capital markets. Common law countries, especially relative to French civil law countries, have stronger investor protections and more developed capital markets.

7.2 Economic intuition

Outside finance requires trust that insiders will not expropriate outside investors. Legal rules give investors control rights and legal claims, while enforcement makes those claims credible. When investor protection is weak, entrepreneurs must rely more on insiders, families, banks, governments, or concentrated owners. When protection is strong, dispersed outside equity and broad public debt markets become more viable.

7.3 Takeaway

The paper's enduring message is that finance is institution-dependent. Capital markets do not grow only because firms need funds; they grow when legal institutions make securities credible claims on cash flows. Investor protection is therefore a foundation of external finance.